Problem Sheet 1. - Acids and Bases

1. Identify each of the following substances as a strong acid, a weak acid, a strong base or a weak base:

nitric acid	ammonia	$Ca(OH)_2$
sodium carbonate	H_2SO_4	phosphoric acid
potassium hydroxide	vinegar	sodium hydrogencarbonate
HCl	KOH	citric acid

2. Complete the gaps:

The theory that an acid is a substance which produces h…… ions in water was first proposed by the Swedish chemist, Svante Arrhenius. Arrhenius proposed that hydrogen ions were produced by the ionisation of an a…….. in water. For example, in hydrochloric acid, the hydrogen chloride molecules are ionised to form h…… ions and c…….. ions: HCl → ……. + ……

Similarly, sulfuric acid and nitric acid i…….. in aqueous solutions to form h…….. ions. Each of these acids is a s……. acid because it is essentially completely ionised in aqueous solution. In hydrochloric acid, for example, there are very few unionised h…….. c……. molecules. Virtually a…….. the molecules have been ionised to form ……. and ……..ions.

In many acids such as e…….. acid and citric acid only a small proportion of the dissolved molecules are ionised. These acids are called w…… acids. The equation for the ionisation of ethanoic acid is as follows: CH_3COOH → …… + ……

A solution of ethanoic acid would therefore contain …… , ……. and unionised ……., together with water.

3. Give the formulae of the particles, apart from water, that would be present in the following solutions. Choose your answers from the formulae given in the box:

H^+	OH^-	HCl	SO_4^{2-}	CH_3COO^-	K^+		NO_3^-
	Br^-	Cl^-	CO_3^{2-}	CH_3COOH	HNO_3	Na^+	

a. a solution of hydrochloric acid

b. a solution of nitric acid

c. a solution of ethanoic acid

d. a solution of sodium hydroxide

4. Give the colour to which you would expect the indicator to change, when added to each of the following aqueous solutions:

Solution	Indicator	
	Phenolphthalein	Litmus
nitric acid calcium hydroxide ammonia H_2SO_4		

5. Identify each of the following statements as true or false. If the statement is false, then rewrite it to make it true.

   a. Ethanoic acid and citric acid are examples of weak acids.

   b. Hydrochloric acid and sodium hydroxide are examples of strong acids.

   c. When an acid dissolves in water, many hydroxide ions are formed.

   d. Nitric acid turns litmus a red colour.

   e. A potassium hydroxide solution will turn litmus a blue colour and phenolphthalein a pink colour.

   f. When hydrogen chloride (HCl) is added to water, only some of the dissolved molecules break up into H^+ and Cl^- ions.

   g. When ammonia is added to water, only some of the dissolved ammonia molecules form OH^- ions.

   h. Carbonic acid is present in "fizzy" drinks.

   i. Orange juice has a pH greater than 7.

   j. Ammonia solution has a pH greater than 7.

   k. A neutral solution has a pH of 7.0.

   l. All acidic solution have pH values less than 7.0.

   m. Sodium carbonate and sodium hydrogencarbonate are examples of weak acids.

6. Match up the correct pH value with the corresponding solution:

a concentrated solution of hydrochloric acid	pH 7.0
a solution of ammonia	pH 1.0
a concentrated solution of sodium hydroxide	pH 3.5
pure water	pH 10.0
vinegar	pH 14.0

7. For the following questions, choose the best alternative:

   a. Ethanoic acid is a weak acid because
   A it is not very soluble in water
   B it is not a very good conductor of electricity
   C it forms only a few hydrogen ions in solution
   D it is produced by living organisms

   b. Which solutions will both turn methyl orange a red colour?
   A sodium hydroxide and ammonia
   B potassium hydroxide and oxalic acid
   C soda water and hydrochloric acid
   D lemon juice and sodium carbonate

   c. Which of the following properties is NOT typical of a base?
   A when dissolved in water, it forms hydroxide ions
   B turns phenolphthalein pink
   C when dissolved in water, it forms a solution with a pH greater than 7
   D it forms hydrogen ions when dissolved in water

   d. Nitrous acid is a weak acid and it has the formula of HNO_2. Apart from water molecules, which particles would you expect to be present in a solution of HNO_2?
   A OH^- and NO
   B only HNO_2
   C H^+ and NO_2^-
   D H^+, NO_2^- and HNO_2

   e. Hydrobromic acid is a strong acid and it has the formula of HBr. Apart from water molecules, which particles would you expect to be present in a solution of hydrobromic acid?
   A OH^- and Br^-
   B only HBr
   C H^+ and Br^-
   D H^+, Br^- and HBr

   f. Barium hydroxide has the formula of $Ba(OH)_2$. Apart from water molecules, which particles would you expect to be present in a solution of barium hydroxide?
   A OH^- and Ba^{2+}
   B only $Ba(OH)_2$
   C H^+ and BaO
   D OH^-, Ba^{2+} and $Ba(OH)_2$

ANSWERS - Acids and Bases Problem Sheets.

Problem Sheet 1.

1. nitric acid - **strong acid** | ammonia - **weak base** | $Ca(OH)_2$ - **strong base**
   sodium carbonate - **weak base** | H_2SO_4 - **strong acid** | phosphoric acid - **weak acid**
   potassium hydroxide - **strong base** | vinegar - **weak acid** | sodium hydrogencarbonate - **weak base**
   HCl - **strong acid** | KOH - **strong base** | citric acid - **weak acid**

2. The theory that an acid is a substance which produces **hydrogen** ions in water was first proposed by the Swedish chemist, Svante Arrhenius. Arrhenius proposed that hydrogen ions were produced by the ionisation of an **acid** in water. For example, in hydrochloric acid, the hydrogen chloride molecules are ionised to form **hydrogen** ions and **chloride** ions: $HCl \rightarrow \mathbf{H^+} + \mathbf{Cl^-}$
   Similarly, sulfuric acid and nitric acid **ionise** in aqueous solutions to form **hydrogen** ions. Each of these acids is a **strong** acid because it is essentially completely ionised in aqueous solution. In hydrochloric acid, for example, there are very few unionised **hydrogen chloride** molecules. Virtually **all** the molecules have been ionised to form **hydrogen** and **chloride** ions.
   In many acids such as **ethanoic** acid and citric acid only a small proportion of the dissolved molecules are ionised. These acids are called **weak** acids. The equation for the ionisation of ethanoic acid is as follows: $CH_3COOH \rightarrow \mathbf{CH_3COO^-} + \mathbf{H^+}$
   A solution of ethanoic acid would therefore contain $\mathbf{H^+}$, $\mathbf{CH_3COO^-}$ and unionised $\mathbf{CH_3COOH}$, together with water.

3. a. H^+ and Cl^- b. H^+ and NO_3^-
   c. CH_3COO^-, H^+ and CH_3COOH d. Na^+ and OH^-

4.

Solution	Indicator	
	Phenolphthalein	Litmus
nitric acid	colourless	red
calcium hydroxide	pink	blue
ammonia	pink	blue
H_2SO_4	colourless	red

5. a. True
   b. False Hydrochloric acid is a strong acid and sodium hydroxide is a **strong base**
   c. False When an acid dissolves in water, many **hydrogen** ions are formed.
   d. True
   e. True
   f. False When hydrogen chloride (HCl) is added to water, **all** of the dissolved molecules break up into H^+ and Cl^- ions.
   g. True
   h. True
   i. False Orange juice has a pH **less** than 7.
   j. True
   k. True
   l. True
   m. False Sodium carbonate and sodium hydrogencarbonate are examples of weak **bases**.

6. a concentrated solution of hydrochloric acid pH 1.0
   a solution of ammonia pH 10.0
   a concentrated solution of sodium hydroxide pH 14.0
   pure water pH 7.0
   vinegar . pH 3.5

7. a. C b. C c. D d. D e. C f. A